Downloaded from specialpapers.gsapubs.org on June 2, 2012

Geological Society of America Special Papers

Paleomagnetic constraints on the evolution of Paleozoic suspect terranes from southern South America

Augusto E. Rapalini, Ricardo A. Astini and Carlos M. Conti

Geological Society of America Special Papers 1999;336;171-182
doi: 10.1130/0-8137-2336-1.171

Email alerting services click www.gsapubs.org/cgi/alerts to receive free e-mail alerts when new articles cite this article

Subscribe click www.gsapubs.org/subscriptions/ to subscribe to Geological Society of America Special Papers

Permission request click http://www.geosociety.org/pubs/copyrt.htm#gsa to contact GSA

Copyright not claimed on content prepared wholly by U.S. government employees within scope of their employment. Individual scientists are hereby granted permission, without fees or further requests to GSA, to use a single figure, a single table, and/or a brief paragraph of text in subsequent works and to make unlimited copies of items in GSA's journals for noncommercial use in classrooms to further education and science. This file may not be posted to any Web site, but authors may post the abstracts only of their articles on their own or their organization's Web site providing the posting includes a reference to the article's full citation. GSA provides this and other forums for the presentation of diverse opinions and positions by scientists worldwide, regardless of their race, citizenship, gender, religion, or political viewpoint. Opinions presented in this publication do not reflect official positions of the Society.

Notes

Geological Society of America

THE GEOLOGICAL SOCIETY OF AMERICA

Downloaded from specialpapers.gsapubs.org on June 2, 2012

Geological Society of America
Special Paper 336
1999

Paleomagnetic constraints on the evolution of Paleozoic suspect terranes from southern South America

Augusto E. Rapalini
Laboratorio de Paleomagnetismo Daniel Valencio, Departamento de Ciencias Geológicas, FCEyN, Universidad de Buenos Aires, Pabellón 2, Ciudad Universidaria, 1428 Buenos Aires, Argentina
Ricardo A. Astini
Cátedra de Estratigrafia y Geología Histórica, Facultad de Ciencias Exactas, Físicas y Naturales, Universidad Nacional de Córdoba, Av. Velez Sarsfield 299, 5000, Córdoba, Argentina
Carlos M. Conti
Laboratorio de Paleomagnetismo Daniel Valencio, Departamento de Ciencias Geológicas, FCEyN, Universidad de Buenos Aires, Pabellón 2, Ciudad Universidaria, 1428 Buenos Aires, Argentina

ABSTRACT

The Paleozoic evolution of southwestern Gondwana apparently involved the accretion of several large suspect terranes at different times and under different tectonic circumstances. The recent acquisition of paleomagnetic data from Paleozoic rocks of this region, albeit still scarce, permit establishment of some constrains on the tectonic and paleogeographic evolution of some of these terranes. A reliable paleomagnetic pole from the Early Cambrian Cerro Totora Formation in northern Precordillera confirms that the Argentine Precordillera is an exotic terrane that was part of or adjacent to Laurentia, very likely at the Ouachita Embayment, in the Early Cambrian. Paleomagnetic poles from an Early Ordovician magmatic belt in northwest Argentina permit the identification of a peri-Gondwanic–rotated terrane, called the Eastern Puna–Famatina terrane, with volcanic arc affinities. Early Paleozoic paleomagnetic poles from the Western Puna (Sierra de Almeida, Chile) were previously interpreted as evidence of a para-autochthonous nature of the Arequipa-Antofalla Massif that rotated successively clockwise and counterclockwise. New preliminary paleomagnetic data in the same region (Salar del Rincón, Argentina) are not easily reconciled with the previous data, unless a Late Ordovician pre-tectonic remagnetization is assumed for the latter. The paleomagnetic and geologic data permit an alternate interpretation based on the fact that the Arequipa and Antofalla blocks can be considered as different terranes. In this case the paleomagnetic data obtained so far is representative of the Antofalla block alone and suggests at least 1,000-km latitudinal displacement of this block with respect to its present position in the South American margin since the Late Cambrian–Early Ordovician. Accretion of this terrane to the Gondwana margin must have occurred in latest Ordovician. Paleozoic paleomagnetic poles from Patagonia indicate that this terrane shares a common apparent polar wander path (APWP) with Gondwana since the Early Devonian. This suggests that no significant movement occurred between these two crustal blocks since that time. Many questions regarding the complex tectonic evolution of southwestern South America that can potentially be addressed by paleomagnetism remain unanswered.

Rapalini, A. E., Astini, R. A., and Conti, C. M., 1999, Paleomagnetic constraints on the evolution of Paleozoic suspect terranes from southern South America, *in* Ramos, V. A., and Keppie, J. D., eds., Laurentia-Gondwana Connections before Pangea: Boulder, Colorado, Geological Society of America Special Paper 336.

Downloaded from specialpapers.gsapubs.org on June 2, 2012

INTRODUCTION

Paleomagnetism has proved to be an essential tool in deciphering the tectonic evolution of continental margins in which accretion of displaced and exotic terranes was an important process (Beck 1989; Van der Voo, 1993). The paleomagnetic technique permits the determination of quantitative paleolatitudinal and paleoazimuthal controls in the evolution of terranes prior to and during their accretion, as well as determination of the age of this process. Sometimes, paleomagnetism can be decisive in ruling out or supporting some tectonic models, and the paleogeography of most exotic terranes is very difficult to ascertain without the aid of paleomagnetic data.

The Paleozoic evolution of the western continental margin of southern South America was a complex one, apparently involving the accretion and displacement of several large terranes in pre–Late Paleozoic times (e.g., Ramos, 1984, 1988; Ramos et al., 1986; Mpodozis and Ramos, 1989; Astini et al., 1995). Although not fully known in detail, the assembly of present-day South America probably began in Late Proterozoic times (Ramos, 1988) when a few cratonic nuclei were assembled into the Gondwana supercontinent (see Rogers et al., 1994). However, large areas of southwestern South America are not part of these more ancient, cratonic blocks. Instead, they were apparently accreted later on, at different times and under different tectonic conditions, displaying a pattern of younger ages of accretion toward the present western continental margin. Although a broad picture of successive accretions during latest Proterozoic-Paleozoic times have been obtained, mainly from geologic and biostratigraphic considerations, the details of this evolution are still poorly known. Probably this has been due in part to the scarcity of paleomagnetic data from Paleozoic rocks in southwestern South America. Yet, in recent years, a few significant paleomagnetic results have been obtained that place important constraints on the tectonic evolution of the southwestern continental margin of Gondwana during the Paleozoic. These results, together with their tectonic and paleogeographic implications, are briefly reviewed in this contribution.

ARGENTINE PRECORDILLERA

The Argentine Precordillera (AP in Fig.1) is the most clearly exotic Paleozoic terrane of this margin (see Astini and Thomas, this volume). Its present morphostructural features are due to Andean tectonics (Jordan et al., 1983). It is a thrust-fold belt located to the east of the main Andes, the uplift of which is due to subduction of the oceanic Nazca plate under the South American continent. The north-south extent of the Paleozoic block, however, probably exceeds 800 km, reaching the San Rafael block in southern Mendoza (Astini et al., 1995; Bordonaro et al., 1996). The Argentine Precordillera consists of a crustal block of Greenvillian age (Mahlburg Kay et al., 1996) with a thick cover of folded Paleozoic and Tertiary rocks (Baldis and Bordonaro, 1981; Allmendinger et al., 1990). The development of a thick carbonate platform between the Early Cambrian and Early Ordovician with typical Laurentian faunas (see Benedetto et al., 1995; Astini et al., 1995) is its most outstanding stratigraphic fea-

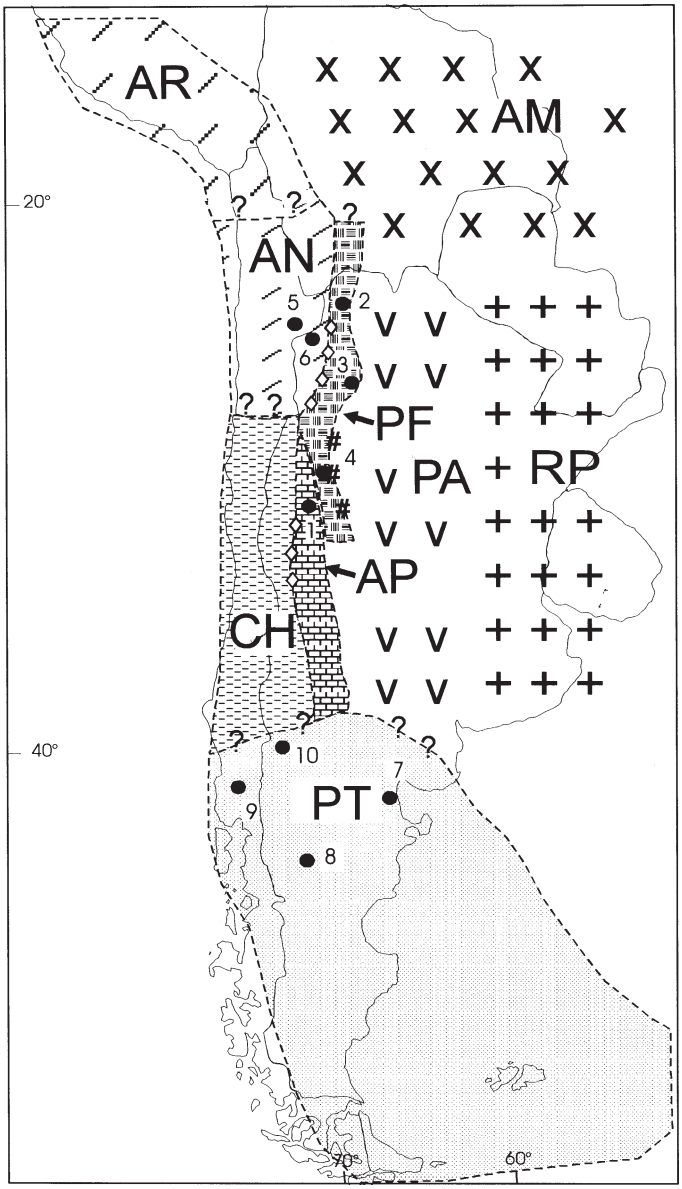


Figure 1. Simplified map of main Late Proterozoic to Paleozoic suspect terranes in southwestern South America. AR = Arequipa Massif; AN = Antofalla block; CH = Chilenia; AP = Argentine Precordillera; PF = Eastern Puna–Famatina; PT = Patagonia; PA = Pampia; RP = Rio de la Plata craton; AM = Amazonia craton. Question marks indicate terrane boundaries with hypothetical locations; open diamonds: exposed ophiolites; # symbols: mylonitic zones; solid circles: location of paleomagnetic studies discussed in the text. Numbers: 1 refers to Cerro Totora; 2, Eastern Puna; 3, Cuchiyaco; 4, Famatina; 5, Sierra de Almeida; 6, Quebrada de Lari; 7, Sierra Grande; 8, Sierra de Tepuel; 9, Chilean granites; 10, Choiyoi volcanics. Simplified from Ramos (1988), Conti et al. (1996), and Bahlburg and Hervé (1997). Minor suspect terranes on the Chilean coast, such as Mejillonia (Bahlburg and Hervé 1997), Pichidangui (Forsythe et al., 1987) and Madre de Dios (Forsythe and Mpodozis, 1979), are not shown.

Downloaded from specialpapers.gsapubs.org on June 2, 2012

ture. Despite this being known for several decades (Poulsen, 1958; Borrello, 1965), it was not until much later (Ramos et al., 1986; Dalla Salda et al., 1992; Astini et al., 1995) that the possibility of the Argentine Precordillera being an accreted terrane was considered seriously.

The 1995 Penrose Conference on the possible Laurentian origin of the Argentine Precordillera (Dalziel et al., 1996) concluded that the Argentine Precordillera is most likely an exotic fragment derived from Laurentia and accreted to Gondwana sometime during the Ordovician. The main evidence for this is the majority of Cambrian fossil faunas nearly identical to those from continental shelf successions of Eastern Laurentia and unique in South America, the very similar Cambrian-Ordovician stratigraphic successions to those of the southern Appalachians, and the isotopic data from the basement rocks with much closer affinities with the Appalachian basement than with adjacent basement rocks in South America. The evolution of the faunal provincialism and the stratigraphy (Benedetto et al., 1995) indicate clear Gondwanan affinities for the Argentine Precordillera by the Late Ordovician. This suggests that the accretion of the Argentine Precordillera took place sometime during the Ordovician. Despite this evidence, no consensus has been reached on the exact timing of separation from Laurentia and accretion to Gondwana, the tectonic mechanism involved in this process or the exact position and kinematics of rifting (Dalla Salda et al., 1992; Astini et al., 1995; Thomas and Astini, 1996; Dalziel, 1997).

Until very recently, no paleomagnetic data were available to test and constrain the tectonic models mentioned before. This was due, in part to the few paleomagnetic studies carried out on this region, but also to an apparently widespread and pervasive Permian remagnetizing event that inhibited determination of the primary magnetic components from many of the Lower Paleozoic rocks (Rapalini and Tarling, 1993; Truco and Rapalini, 1996).

A successful paleomagnetic study was recently carried out on the Lower Cambrian Cerro Totora Formation (Rapalini and Astini, 1998), exposed in the northern Precordillera (location 1 in Fig. 1). The Cerro Totora Formation consists in a mixed succession of evaporites, red siliciclastics, and carbonates (Astini and Vaccari, 1996) with trilobites of the Bonnia-Olenellus zone (Vaccari, 1990). This succession has been interpreted as typical synrift deposits (Astini and Vaccari, 1996), possibly accumulated during the initial rifting of the Argentine Precordillera from the Ouachita embayment of south-central North America (Thomas and Astini, 1996).

Rapalini and Astini (1998) reported a paleomagnetic study from 12 sites (80 samples) located on red siltstones (9 sites), dolomitic limestones (2 sites), and green shales (1 site) of the Cerro Totora Formation. Following standard demagnetization techniques within-site consistent characteristic remanent magnetization directions were isolated from the nine hematite-bearing siltstone sites and one site of the magnetite-bearing dolomitic limestone. A positive fold test indicated a pre-tectonic (pre–Late Tertiary) magnetization. A virtual geomagnetic pole was computed for each site (N = 10), the mean of which is the paleomagnetic pole of the Cerro Totora Formation (CT in Table 1). The position of CT is not consistent with any expected pole position for South America in the Phanerozoic, suggesting that a post-Cambrian remagnetization of the rocks is unlikely. CT also disagrees with the Cambrian path for the already assembled Gondwana (Fig. 2A) (Meert and Van der Voo, 1996). However, if the Argentine Precordillera is placed in the Ouachita embayment, according to the hypothesis of Thomas and Astini (1996), CT becomes perfectly consistent in position and age with the Cambrian segment of the Laurentian APWP (Fig. 2B).

This paleomagnetic result provides strong evidence that the Argentine Precordillera is an exotic terrane derived from Laurentia and most likely from the southern Appalachians. This also implies that all early Paleozoic or older terranes located to the west (i.e., the basement of Chilenia) (Ramos et al., 1986) are necessarily allochthonous. Since there is no younger early Paleozoic paleomagnetic pole yet available for the Argentine Precordillera, its kinematic history between the Early Cambrian and the Late Ordovician remains unconstrained by paleomagnetic data.

EASTERN PUNA–FAMATINA TERRANE

An outstanding geologic feature of northwest Argentina is a north-trending belt of Lower Ordovician magmatic rocks (Fig.1) that comprises the Famatina system and the “Faja Eruptiva” of Eastern Puna (Puna Oriental). The age of this belt is very well defined by means of fossil assemblages found in intercalated sediments as Tremadoc–early Llanvirn (e.g., Coira, 1973; Koukharsky and Mirré, 1974; Coira and Koukharsky, 1991; Aceñolaza, 1992; Vaccari et al., 1992), as well as radiometric datings on intrusive bodies in the Eastern Puna (Omarini et al., 1984; Lork and Bahlburg, 1993) between 476 and 467 Ma. This volcanism has been assigned to a magmatic arc on the basis of its geochemical signature and geotectonic setting (Coira et al., 1982; Ramos, 1986; Mannheim, 1993; Toselli et al., 1996). This volcanic arc was apparently emplaced on oceanic or quasi-oceanic crust in northern Puna (Ramos, 1986) and Famatina (Toselli et al., 1996), whereas in southern Puna there is evidence of underlying continental crust (Ramos, 1986).

Conti et al. (1996) presented paleomagnetic data from Lower Ordovician rocks at four localities (16 sites) on this belt, including a reassessment of previously published data by Valencio et al. (1980) on the Suri Formation in the Famatina system (locations 2–4 in Fig.1). Conti et al. (1996) found consistent pretectonic magnetizations on different units of the same age along this belt, and computed three paleomagnetic poles of Early Ordovician age for the Puna Oriental (Eastern Puna), Cafayate area (Cuchiyaco Granodiorite), and the Famatina system (Table 1). The same magnetization direction was found in different lithologies carried by different magnetic minerals. The poles derived from the paleomagnetic directions are consistent between them, but disagree with the Early Ordovician mean pole position for Gondwana (Grunow, 1995). The disagreement is found in

Downloaded from specialpapers.gsapubs.org on June 2, 2012

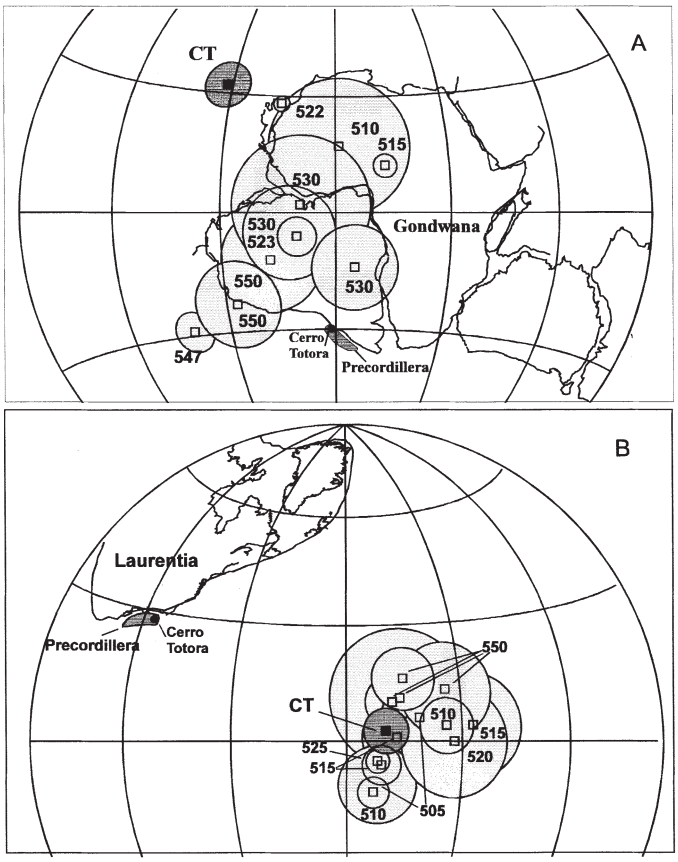


Figure 2. A, Early Cambrian paleomagnetic pole from the Cerro Totora Formation (CT) in African coordinates after Gondwana reconstruction (de Wit et al., 1988) and the Gondwana 550–510-Ma APWP (Meert and Van der Voo, 1996). The Precordillera has been kept attached to South America in its present location. Numbers indicate the most likely age (in Ma) of the Gondwana poles. B, Position of Cerro Totora Formation pole (CT), after placing the Argentine Precordillera against the southeastern Laurentian margin following the model of Thomas and Astini (1996), and the 550–505 APWP of Laurentia (Torsvik et al., 1996). Numbers indicate the most likely age (in Ma) of each Laurentian paleomagnetic pole. From Rapalini and Astini (1998).

declination only (Fig. 3), with no paleolatitudinal discordance. A clockwise rotation of 52.6° ± 11.1° was computed by Conti et al. (1996). These paleomagnetic results led these authors to propose that the Lower Ordovician volcano-sedimentary belt extending from La Rioja province (Famatina) to Jujuy province (eastern Puna), and called Puna Oriental–Famatina (PF in Fig. 1) is a rotated terrane that was accreted to Gondwana in Middle Ordovician times. Geologic data (Coira et al., 1982; Allmendinger et al., 1983; Astini et al., 1995, Blasco et al., 1996; Bahlburg and Hervé, 1997) support the existence of an oceanic realm between the Arequipa-Antofalla and the Argentine Precordillera terranes on one side and Puna Oriental–Famatina on the other during Ordovician times. Conversely, the eastern boundary of this terrane is clearly defined by a thick mylonitic belt (Lopez and Toselli, 1993) in the south that marks the limit with the Proterozoic Pampia terrane (PA in Fig. 1). The northeastern boundary is not so clearly defined by a mylonitic zone, as by the tectonic discontinuity between Puna and Cordillera Oriental provinces (Allmendinger et al., 1983; Mon and Hongn, 1987; Bahlburg and Hervé, 1997). The age of rotation and accretion cannot be determined on the basis of the available paleomagnetic data; however, the most likely age on geologic considerations corresponds to the Middle Ordovician, both in Puna Oriental (Bahlburg and Hervé, 1997) and in Famatina (Rapela et al., 1992; Toselli et al., 1996).

The similar paleolatitudes indicated by the paleomagnetic data for this terrane and the southwestern continental margin of Gondwana in Early Ordovician times suggest that the Puna Oriental–Famatina was a peri-Gondwanic magmatic arc. A similar paleogeographic reconstruction was obtained for the Famatina system by Benedetto and Sánchez (1996) and Astini and Benedetto (1996) on the basis of biogeographic considerations, and by Toselli et al. (1996) based on the geochemical signature of the magmatic rocks.

The Puna Oriental–Famatina volcanic arc was probably only one of several short-lived magmatic arcs that developed in the southern Iapetus during Ordovician time (Van der Voo, 1993; Mac Niocaill et al., 1997; Dalziel, 1997). According to Astini et al. (1995), the Famatina volcanic arc (and its probable northward extension in eastern Puna) was produced by the consumption of oceanic crust associated with the approach of the Precordillera terrane toward the Gondwana margin. A rotation of this arc in Middle Ordovician time produced its accretion to the Gondwana margin.

WESTERN PUNA (AREQUIPA?–ANTOFALLA) TERRANE

The existence of Precambrian crystalline rocks along the southern coast of Perú, generally referred to as the Arequipa Massif (AR in Fig. 1), has long posed the possibility of an exotic origin. The rocks belonging to this massif gave Rb/Sr radiometric ages of 1,910 ± 36 and 1,918 ± 33 Ma (Dalmayrac et al., 1977; Shackleton et al., 1979). This Proterozoic block was extended southward in many models to include outcrops of metamorphic rocks in Antofalla and Belén in northern Chile, with ages ranging from 1,460 to 777 Ma (Pacci et al., 1980; Damm et al., 1990), leading to a larger Arequipa-Antofalla block (Ramos, 1988) (AR-AN in Fig. 1). Different early Paleozoic tectonic evolutions (see Forsythe et al., 1993) have been inferred for the northern (Arequipa) and the southern (Antofalla) parts of this terrane. In the north, no evidence has been found for the existence of ocean floor of early Paleozoic age between the Precambrian coastal rocks and the basement of the Amazon basin (Dalmayrac et al., 1980; Sempere, 1995). However, Ordovician ophiolites exist in the Western Puna of Argentina to the east of the Antofalla metamorphic rocks (Allmendinger et al., 1983; Blasco et al., 1996), suggesting the existence of an ocean between this area and Gondwana in Ordovician time. Recent radiometric datings by Basei et al.(1996) cast

Downloaded from specialpapers.gsapubs.org on June 2, 2012

TABLE 1. SUMMARY OF PALEOMAGNETIC POLES FROM PALEOZOIC SUSPECT TERRANES OF SOUTHWESTERN SOUTH AMERICA

Terrane	Formation	Paleomagnetic Pole (PP) Latitude (°)	Longitude (°)	A95 (dp/dm) (°)	Paleomagnetic Results*	PP Age	Ref†
Precordillera	Cerro Totora F. (CT)	37.0	314.1	5.8	AF/Th demag, +FT N = 10 (n = 65), IRM	Early Cambrian	1
Puna Oriental-Famatina	Acoyte F., Chiquero F., (PO)	-17.7	357.9	13.0	AF/Th demag, + FT, IRM, N = 5 (n = 21).	Early Ordovician	2
	Cuchiyaco Gr. (CY)	-30.0	0.5	10.0	AF/Th demag, IRM, N = 5 (n = 23)	Early Ordovician	2
	Suri F. (FS)	-13.0	356.3	6.0	AF/Th demag, N = 6 (n = 18)	Early Ordovician	2
	Overall mean (PF)	-21.8	358.7	7.5	+FT, Consistency test N = 16 (n = 61)	Early Ordovician	2
Western Puna	Choschas Pluton + Pre-Sil lavas (CHO)	58.1	309.2	(3.8/7.5)	AF/Th demag, both polarities, n = 24	Late Cambrian–Early Ordovician	3
	Late Ordovician plutons (OP)	0.0	271.0	(12.4/13.7)	AF/Th demag, +FT, both polarities, n = 20	Late Ordovician	3
	Pre-Sil lavas (roof pendant?) (OL)	-25.2	260.5	(13.7/15.0)	AF/Th demag, both polarities, n = 16	Silurian?	3
	Las Vicuñas F. (FV)	-45.1	283.4	10.6	AF/Th demag, +FT, N = 4 (n = 12)	Late Ordovician?	4
Patagonia	Sierra Grande F. (lower member) (SG1)	3.4	238.0	(8.0/14.0)	AF/Th demag, both polarities, N = 4 (n = 14)	Silurian??	5
	Sierra Grande F. (upper member) (SG2)	-42.0	283.5	20.0	Th/Ch demag, +FT, N= 2 (n = 10)	Early Devonian	5
	Tepuel Group (TP)	-31.7	316.1	(15.0/16.0)	AF/Th demag, +FT, IRM, detrital rem, N = 16 (n = 62)	Middle Carboniferous	6
	Chilean Intrusives (CI)	-57.4	323.5	19.0	AF/Th demag, both polarities, N = 7 (n = 58)	Late Carboniferous–Early Permian	7
	Sierra Grande F. (SG3)	-77.3	310.7	(7.0/8.0)	AF/Th demag, IRM, FT syntectonic, N = 13 (n = 88)	Late Permian?	8
	Choiyoi F. (CH)	-21.0	232.3	8.0	AF/Th demag, +FT, N = 33 (n = 85)	Late Permian	9

*AF/Th/Ch demag mean complete demagnetization by alternating field, thermal cleaning, or chemical leaching; +FT = positive fold test; N = number of sites; n = number of samples; IRM = isothermal remanence experiments..
†1 = Rapalini and Astini, 1998; 2 = Conti et al., 1996; 3 = Forsythe et al., 1993; 4 = this chapter; 5 = Rapalini and Vilas, 1991; 6 = Rapalini et al., 1994; 7 = Beck et al., 1991; 8 = Rapalini, 1998; 9 = Rapalini et al., 1989.

doubts on the Proterozoic ages of some of the Chilean (Antofalla) outcrops. This, as well as evidence for substantial differences in age and composition of the basement of the Andes to the north and south of 20°–21°S (Wörner et al., 1992; Aitchenson et al., 1995), led Bahlburg and Hervé (1997) to postulate that the Arequipa-Antofalla block be subdivided into two blocks: an Early to Middle Proterozoic Arequipa block in the north and a Late Proterozoic–Early Paleozoic Antofalla block in the south (Fig. 1).

Forsythe et al. (1993) recently presented the first early Paleozoic paleomagnetic data from the western Puna in the Sierra de Almeida (northern Chile) (location 5 in Fig. 1). These data are summarized in Table 1 (poles CHO, OP, and OL). The lower Paleozoic rocks they studied belong to a magmatic belt known as the Faja Eruptiva de la Puna Occidental (Western Puna Eruptive belt) (Palma et al., 1986). Three paleomagnetic poles were obtained from these rocks. The oldest one corresponds to the Late Cambrian (502 ± 10 Ma) Choschas Granite and the Cambrian-Ordovician volcanics of the Cordon de Lila Complex (CHO in Fig. 6A,B). This pole presented an inconclusive tilt test but has no resemblance to expected younger poles for Gondwana. A second pole (OP in Table 1 and Fig. 6) was obtained from the Tucucaro, Tilopozo, and Alto del Inca plutons with Rb/Sr ages of 441 ± 8, 452 ± 4, and 468 ± 100, 433 ± 22 Ma, respectively. The remanence directions from these plutons were corrected for tilt of a pre-Devonian unconformity, resulting in a positive tilt test. A third pole (OL in Table 1 and Fig. 6) was obtained for the Cambrian-Ordovician lavas of the Cordón de Lila Complex at a different locality where they may be a roof pendant of a Silurian pluton. Forsythe et al. (1993) suggested a remagnetized Silurian age for this pole.

Downloaded from specialpapers.gsapubs.org on June 2, 2012

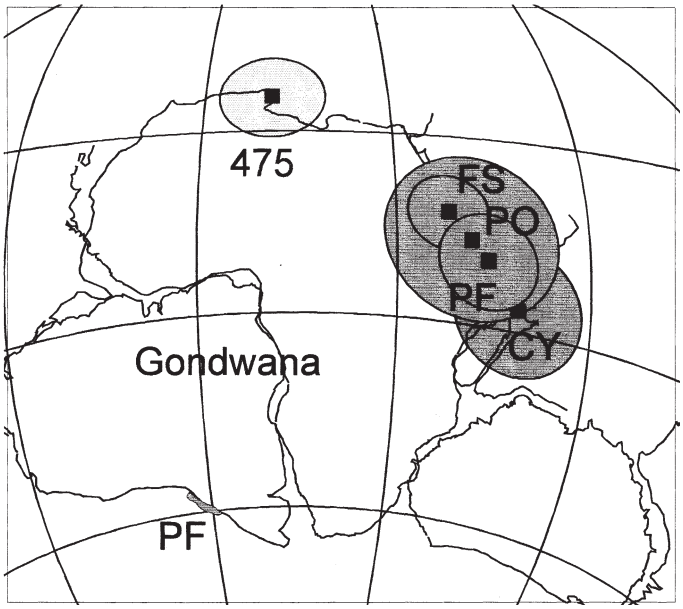


Figure 3. Comparison of the Early Ordovician paleomagnetic poles from the Eastern Puna–Famatina terrane with the 475-Ma mean paleomagnetic pole for Gondwana (Grunow, 1995) in a Gondwana reconstruction (de Wit et al., 1988). Details and symbols of paleomagnetic poles as in Table 1. PF = Eastern Puna–Famatina terrane. Modified from Conti et al. (1996).

A new paleomagnetic study was carried out recently by Conti (1995) on the Lower Tremadocian Las Vicuñas Formation (Moya et al., 1993). This is a 200-m-thick volcaniclastic sequence exposed at Quebrada de Lari (location 6 in Figs. 1 and 4) in the western Puna of Argentina. The outcrops of this unit are relatively near the Sierra de Almeida in Chile (see Fig. 1) and also belong to the Western Puna Eruptive belt. Thirty-one samples from seven sites were collected from the Las Vicuñas Formation. Application of thermal demagnetization proved ineffective to isolate the magnetic components because of chemical changes caused by the experimental heating. However, AF demagnetization (Fig. 5) permitted the definition of the characteristic remanent magnetization in 12 samples from four sites, presumably carried by magnetite. This remanence was generally isolated at demagnetizing fields ranging from 10 to 80 mT. A secondary component coincident with the present dipole direction was found also in most samples carried by goethite (Conti, 1995). Comparison of remanence directions (Table 2) in situ and after tectonic correction shows a better grouping after full tectonic correction. The pre-Silurian rocks of the area were subjected to at least two main deformational phases: the Late Ordovician Ocloyic and the Late Cenozoic Andean (Moya et al., 1993). The latter is represented by open northwest-trending folds. All sampling sites were located on one limb of the Andean fold (Fig. 4); therefore, the positive result of the fold test suggests a pre-Ocloyic magnetization. However, the small number of samples and the fact that the site with a substantial different structural attitude is represented by a single sample make interpretation of results highly speculative. Nevertheless, the in situ remanence directions do not correspond to those expected for Recent times, which is taken as strong evidence of at least a pre-Andean magnetization. A preliminary paleomagnetic pole for the Vicuñas Formation (FV) is shown in Figure 6 (see also Tables 1 and 2).

Forsythe et al. (1993) interpreted their data from Sierra de

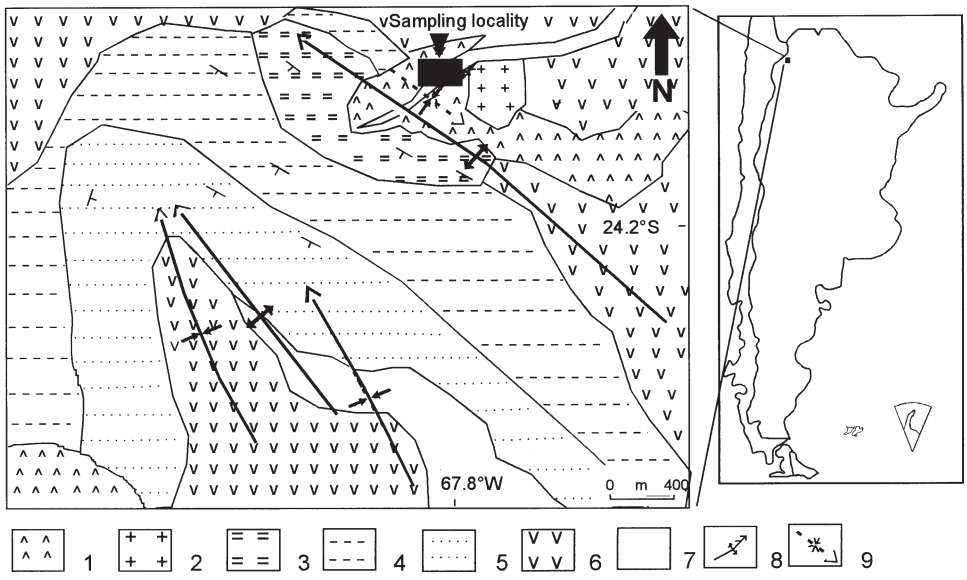


Figure 4. Geologic map of the Quebrada de Lari near Salar del Rincón (western Puna) and location of the sampling locality of the Early Ordovician Vicuñas Formation. Numbers: 1, Vicuñas Formation (Tremadoc); 2, rhyolitic porphyry (Ordovician?); 3, Salar del Rincón Formation (Silurian); 4, Cerro Oscuro Formation (Carboniferous); 5, Arizaro Formation (Permian); 6, Cenozoic volcanics; 7, Recent sediments; 8, Tertiary fold; 9, pre-Silurian fold. Taken from Moya et al. (1993).

Downloaded from specialpapers.gsapubs.org on June 2, 2012

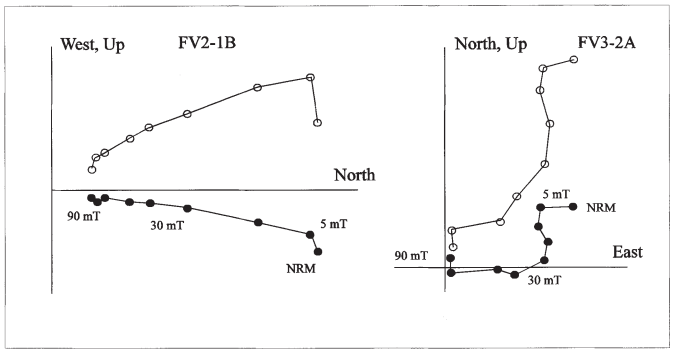


Figure 5. Characteristic magnetic behavior of samples from the Las Vicuñas Formation submitted to AF demagnetization. Closed (open) symbols correspond to vectorial representations on the horizontal (vertical) planes.

Almeida as representative of the entire Arequipa-Antofalla block. It is clear from Figure 6 that some of the poles are not consistent with coeval poles from Gondwana. In particular, the Late Cambrian–Early Ordovician pole (CHO) shows an important departure from the expected position according to the recently presented Early Ordovician mean pole for Gondwana (Grunow, 1995). Forsythe et al. (1993) proposed a counterclockwise rotation of around 50° for the Arequipa-Antofalla block after the Early Ordovician to account for this discrepancy. They also suggested no significant discordance in paleolatitude. The other two poles (OP and OL) were assigned to the Late Ordovician and Early Silurian, respectively. Due to the fact that the Middle Ordovician–Early Devonian segment of the Gondwana APWP is ill-defined (Van der Voo, 1993), Forsythe et al. (1993) suggested that the only significant discordance of the paleomagnetic poles from the Sierra de Almeida with respect to the Gondwana path was for the Late Cambrian–Early Ordovician. These data were used to propose a model of a para-autochthonous evolution of the Arequipa-Antofalla block, in which this block rotated clockwise, probably in the latest Proterozoic, around an Euler pole located in northern Perú. This would lead to the opening of a basin with development of oceanic crust in the southern part, consistent with the geologic observations of ophiolitic rocks only in northwest Argentina. Closure of the basin could not be deciphered from the paleomagnetic data due to difficulty in determining whether the Late Ordovician and Silurian poles were discordant compared with the Gondwana APWP. Geologic considerations pointed to a Late Ordovician (Ocloyic) closure of the basin.

If the position of the Late Cambrian–Early Ordovician (CHO) pole is compared rigorously respect to the ~475-Ma mean pole for Gondwana (Grunow, 1995), it shows significant discordance both in paleoazimuth and paleolatitude. The corresponding values are: for rotation, 49.4° ± 8.6° (counterclockwise), for paleolatitude discordance, 17.7° ± 8.1°. This suggests at least 10° lower paleolatitude for the Sierra de Almeida in Early Ordovician times than expected. Unless this is considered as an artifact of the paleomagnetic data, these data suggest that the Sierra de Almeida is part of a displaced terrane. Although the model presented by Forsythe et al. (1993) can still be sustained, mainly on geologic grounds, an alternate interpretation of the paleomagnetic data is possible.

Considering that the Antofalla block is possibly a separate terrane from the Arequipa Massif (Bahlburgh and Hervé, 1997), the paleomagnetic data obtained by Forsythe et al. (1993) and Conti (1995) correspond only to the Antofalla block. This block apparently has much younger crust (Neoproterozoic?) than the Arequipa Massif and is bounded to the east by Ordovician ophiolitic rocks with mid-ocean ridge basalt geochemical signature (Blasco et al., 1996). This is consistent with the consumption of oceanic crust during Ordovician times to the east of the Antofalla block. These observations suggest that the Antofalla block had an independent evolution from the Arequipa Massif in early Paleozoic times. The paleomagnetic data (CHO pole) suggest that in the Early Ordovician it was located at least 1,000 km (in lower paleolatitudes) from its present position on the South American margin. During Ordovician time this terrane moved toward the continental margin and was accreted in the latest Ordovician–Early Silurian, causing the Ocloyic tectonic event (Bahlburg and Hervé, 1997).

The ill-defined nature of the Late Ordovician–Silurian segment of the Gondwana APWP does not permit the exact time of accretion of the Antofalla block to be determined from the paleomagnetic observations. However, the OP pole, of probable Late Ordovician age, falls west (present-day coordinates) from the generally assumed Middle–Late Ordovician APWP segment of Gondwana. This might suggest that some relative displacement

TABLE 2. PALEOMAGNETIC DATA FROM LAS VICUNAS FORMATION

Site	n	Dec. in Situ (°)	Inc. in Situ (°)	α95 (°)	Bed. Str. (°)	Bed. Dip (°)	Dec. Corr. (°)	Inc. Corr (°)	α95 (°)
FV2	2	57.0	-39.5	.....	341	47	354.7	-78.9	
FV3	5	61.0	-42.7	17.0	338	42	22.6	-82.8	
FV4	4	49.0	-44.8	15.0	338	42	344.7	-76.6	
FV6	1	46.5	-7.6	.....	327	70	16.2	-74.0	
All sites	4 (12)	52.9	-34.0	21.2			3.3	-78.5	5.9

Dec. = declination; Inc. = inclination; Bed. Str. = bedding strike; Bed. dip = bedding dip; Dec. Corr. = declination corrected for bedding; Inc. Corr. = inclination corrected for bedding.

Downloaded from specialpapers.gsapubs.org on June 2, 2012

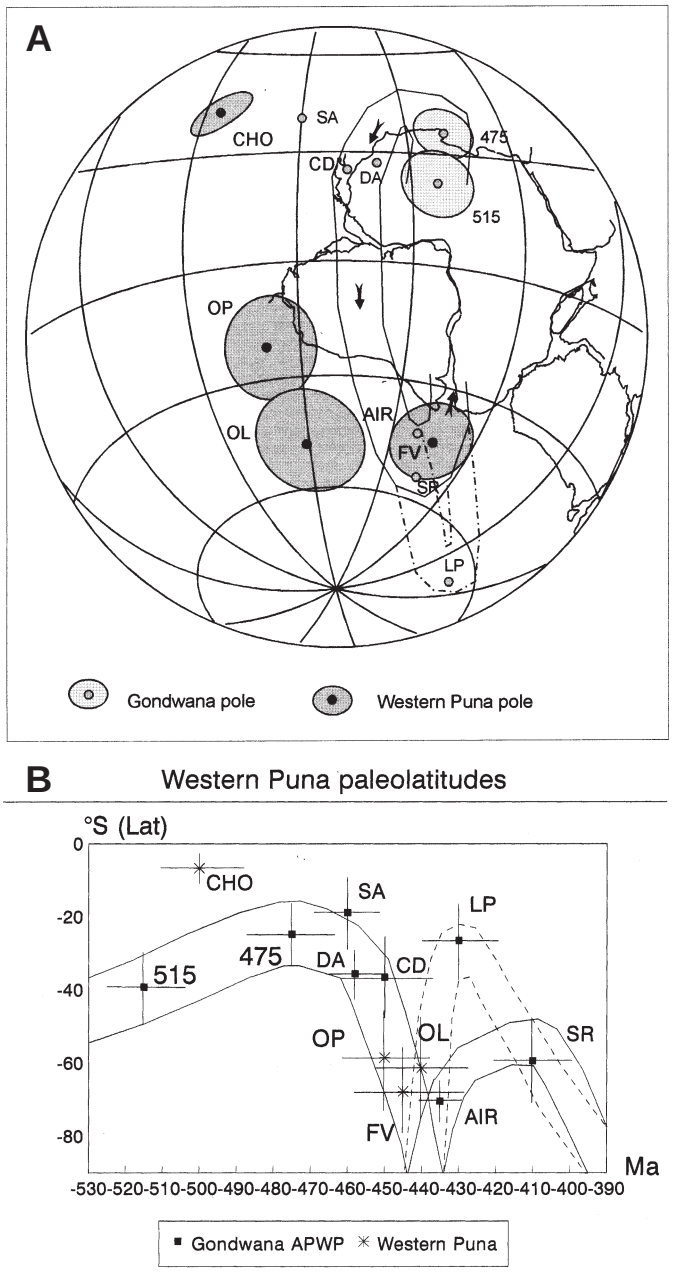


Figure 6. A, Early Paleozoic paleomagnetic poles for the Western Puna and their 95% confidence ovals in a Gondwana reconstruction (according to de Wit et al., 1988). The 515- and 475-Ma mean paleomagnetic poles for Gondwana (Grunow, 1995) and the hypothetical trace of the Ordovician–Devonian apparent polar wander path are also shown. An alternate Silurian segment according to the Lipeon pole (Conti et al., 1995) is shown as dotted lines. Individual poles for Gondwana (SA = Salala Ring Complex; CD = Cedarberg Formation; DA = Damara Granites; AIR = Air Ring Complex; SR = Snowy River Volcanics; from selections by Grunow, 1995, and Conti et al., 1995) are shown without their confidence circles. Details of Western Puna poles are presented in Table 1. B, Paleolatitude vs. geologic time plot for a site in the Western Puna (24°S, 68°W) according to the Gondwana and Western Puna paleomagnetic poles shown in (A) with their estimated errors. Note the discordant paleolatitude obtained from the Choschas pole with respect to the expected paleolatitude from the Gondwana poles. Both possible paleolatitudinal tracks for the Silurian are shown. Letter symbols as in Figure 6A.

(rotation?) took place between the Antofalla and Gondwana land masses in the Late Ordovician. The position of the FV pole is problematic if a Tremadocian age is assumed for it, as it is totally inconsistent with the CHO pole from Sierra de Almeida. The fact that FV is compatible with a latest Ordovician–Early Silurian pole position for Gondwana (Van der Voo, 1993; Conti et al., 1995; Berquó and Ernesto, 1997) suggests that its magnetization is of that age and that it could be related with the beginning of the Ocloyic orogeny.

Another argument presented by Forsythe et al. (1993) in favor of the para-autochthonous nature of the Arequipa-Antofalla block is that the early Paleozoic poles obtained in their study show a trend similar to that of the Gondwana APWP for those times, suggesting that both landmasses moved together. However, Van der Voo (1994) has noted that similar trends are found for various continents for this time interval. He suggested that the "loop" of the Gondwana APWP is due to an episode of true polar wander. In such a case, similar trends would be expected for all continental blocks, regardless of their respective drift histories.

Tosdal et al. (1994) presented isotopic data that suggest that the Arequipa Massif is closely related to the Amazonia craton. These data were obtained from outcrops north of 20°S and therefore pertain to the Arequipa block. The paleomagnetic data, however, were obtained from rocks south of 23°S. Therefore, the boundary between the Arequipa and Antofalla blocks should lie somewhere between 20° and 23°S. It is interesting that a clear change in the character of the Andean basement was found at around 20°–21°S by Wörner et al. (1992) and Aitchenson et al. (1995) from the geochemistry of Cenozoic lavas. Furthermore, Götze et al. (1994) have recently found a major gravimetric high in western Puna extending south of 22°S, which they related to a fundamental change in the basement characteristics at that latitude. They also reported a much higher rigidity of the basement between 20° and 22°S with respect to that between 22° and 26°S.

The hypothesis suggested above should be considered as an alternate interpretation to that presented by Forsythe et al. (1993). However, new and more reliable early Paleozoic paleomagnetic data from both the Arequipa and the Antofalla blocks are needed to determine which, if either, of these hypotheses is correct.

PATAGONIA

Patagonia is the largest of the Paleozoic suspect terranes of southern South America (Fig.1). In some models this terrane has been portrayed as a single tectonic block comprising the Malvinas Plateau (Ramos, 1988; Mahlburgh Kay et al., 1989). The peculiar geologic features of Patagonia have attracted the attention of many geologists since the first decades of this century (Keidel, 1925; Windhausen, 1931). In the last two decades, several hypotheses have proposed an allochthonous nature for Patagonia (Dalmayrac et al., 1980; Martinez, 1980; Ramos, 1984, 1988; Sellés Martinez, 1988). Instead, other models suggest an essentially authochtonous character of this block, at

Downloaded from specialpapers.gsapubs.org on June 2, 2012

least since the early Paleozoic (Varela et al., 1991; Dalla Salda et al., 1992).

The most favored of the "allochthonous" models that considers Patagonia as a single block is that proposed by Ramos (1984). This model suggests that an ocean was consumed between the northern boundary of Patagonia and southern Gondwana by subduction beneath Patagonia until a collision took place in the Middle or Late Paleozoic. Deformation in Sierra de la Ventana area was interpreted as the result of this collision. This model has remained controversial for several reasons: (1) there is a lack of a suture with obducted ocean floor between both continental blocks; (2) there is no biostratigraphic or palaeoclimatic evidence suggesting different palaeogeographies for Patagonia and South America in the Late Palaeozoic; (3) there are comparable Late Proterozoic to early Paleozoic basement rocks in northern Patagonia and the Pampean Ranges (PA in Fig. 1) in central Argentina (e.g., Varela et al., 1991); and (4) the location of the hypothetical boundary in western Argentina and Chile is unclear.

It is clear that high-quality palaeomagnetic data from Paleozoic rocks in Patagonia could solve the controversy surrounding its tectonic history. However, suitable Paleozoic sedimentary or volcanic exposures are scarce in Patagonia with most rocks of this age being intrusive or metamorphic basement. Nevertheless, some paleomagnetic data have been obtained from Paleozoic rocks of Patagonia in recent years (Rapalini et al., 1989, 1994; Beck et al., 1991; Rapalini and Vilas, 1991; Rapalini, 1998). The paleomagnetic poles obtained from Paleozoic rocks in Patagonia are summarized in Table 1. Six paleomagnetic poles of possible Paleozoic age from Patagonia have been published. Three of them (SG1, SG2, SG3) correspond to the Silurian–Early Devonian Sierra Grande Formation, exposed in northeast Patagonia (location 7 in Fig. 1). Another pole (TP) belongs to the middle–late Carboniferous Tepuel Group in central western Patagonia (location 8 in Fig. 1). The CI pole was determined from late Carboniferous–Early Permian (305–285 Ma) intrusive rocks exposed in the Chilean forearc (location 9 in Fig. 1). The CH pole was determined from Late Permian volcanics exposed in northwest Patagonia (location 10 in Fig. 1). Rapalini (1998) has recently discussed the reliability of these poles. The TP and CH poles were considered highly reliable; SG2, SG3, and CI, moderately reliable; while SG1 was qualified as being of low reliability.

The positions of these poles with respect to the middle–late Paleozoic APWP of Gondwana is presented in Figure 7. Poles SG2 of Early Devonian age, TP of middle Carboniferous age, and CI of late Carboniferous–Early Permian age are consistent with the Gondwana path. Pole SG3 is a recently obtained pole (Rapalini, 1998) for the deformation of the Sierra Grande sedimentary rocks. The position is consistent with the Gondwana path for late Early to Late Permian times, suggesting folding and remagnetization of these rocks at that time. The discordant position of CH is due to a local tectonic rotation of the sampling area (Rapalini et al., 1989). CH shows, however, concordant paleolatitudes with that expected from Late Permian poles for Gondwana. The single discordant pole from Patagonia is SG1. While this could be taken as evidence for an accretional origin of Patagonia, the low reliability of this pole suggests that no tectonic interpretation should be based on it (Rapalini, 1998). All poles other than SG1 indicate that both Patagonia and Gondwana shared the same apparent polar wander path since the Early Devonian, suggesting no significant movement of Patagonia with respect to Gondwana since that time.

The paleomagnetic data summarized above suggest that, if Patagonia is an accreted terrane, its collision with Gondwana occurred before the Devonian. However, both the Sierra Grande sediments in northeast Patagonia (Rapalini, 1998) and the sedimentary successions exposed at Sierra de la Ventana, north of the Patagonian margin (Japas, 1987; von Gosen et al., 1990; Cobbold et al., 1991; Tomezzoli and Vilas, 1996) were deformed in Permian times. This renders unlikely any tectonic model involving a continent-continent collision to account for the deformational event around the northern boundary of Patagonia, unless deformation can be explained occurring well over 100 Ma after collision started. Tectonic models that propose that Patagonia was part of Gondwana since the early Paleozoic (e.g., Varela et al., 1991; Dalla Salda et al., 1992), seem to be more consistent with the paleomagnetic data obtained to date. However, determination of the pre-Devonian paleogeographic evolution of Patagonia will require acquisition of new paleomagnetic poles.

CONCLUDING REMARKS

The tectonic evolution of southern South America in the

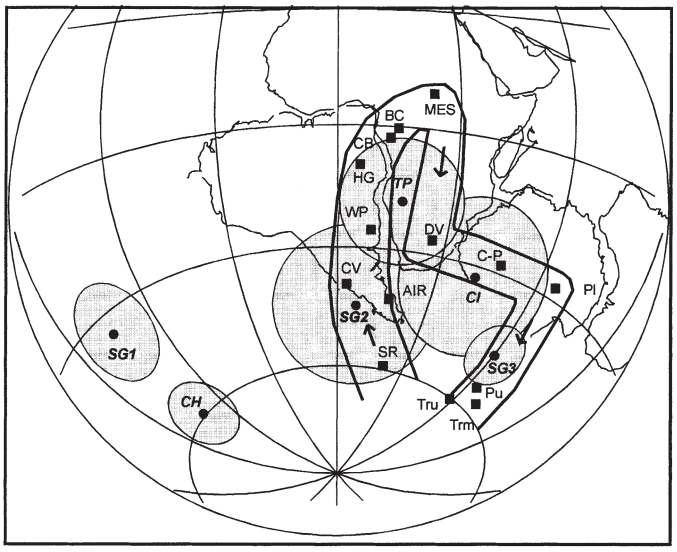


Figure 7. Paleozoic paleomagnetic poles from Patagonia with their 95% confidence ovals and the Early Devonian–Triassic apparent polar wander path of Gondwana. The Gondwana path is based on individual high-quality poles from Early Devonian to Late Carboniferous (see Rapalini, 1998) and mean Western Gondwana poles from Late Carboniferous to Triassic-Jurassic boundary (Van der Voo, 1993). See discussion in the text. Modified from Rapalini (1998).

Downloaded from specialpapers.gsapubs.org on June 2, 2012

Paleozoic probably involved the accretion of several terranes. The recent acquisition of paleomagnetic data on Paleozoic rocks from this region, albeit still scarce, provides constraints on the tectonic and paleogeographic evolution of some of these terranes.

Recently obtained paleomagnetic results from Early Cambrian sediments of the Argentine Precordillera confirm that this terrane was part of Laurentia at that time. Although not constrained by paleomagnetism, it is very likely that by the late Ordovician it was already accreted to the western Gondwana margin based on paleobiogeographic affinities (Benedetto et al., 1995) and continuity of the Gondwanan Hirnantian glacial event throughout the proto-Andean region of South America (Buggisch and Astini, 1993). This implies that all pre-Silurian rocks situated to the west of the Argentine Precordillera are also allochthonous.

Paleomagnetic data from geologic units of Early Ordovician age along a magmatic belt of this age in northwest Argentina led to the definition of an accreted terrane with volcanic arc characteristics called the Puna Oriental–Famatina terrane. This terrane was most likely a peri-Gondwanic, short-lived volcanic arc that was accreted to the Gondwana margin in Middle Ordovician times.

Paleomagnetic results from early Paleozoic rocks in the western Puna magmatic belt were previously interpreted as representative of a para-authochtonous single Arequipa-Antofalla block. This block was supposed to rotate successively clockwise and counterclockwise to rift apart from and then collide with the Gondwana margin. The paleomagnetic and geologic data permit an alternate interpretation based on the fact that the Arequipa and Antofalla blocks can be considered separate terranes. In this case, the paleomagnetic data obtained so far is representative of the Antofalla block alone and suggests at least 1,000 km of latitudinal displacement of this block with respect to its present position on the South American margin for the Late Cambrian–Early Ordovician. Accretion of this terrane to the Gondwana margin by latest Ordovician is consistent with both paleomagnetic and geologic data.

Paleomagnetic poles of different qualities from Patagonia indicate that this terrane shares a common apparent polar wander path with Gondwana since the Early Devonian. This suggests that no significant relative movements occurred between these two crustal blocks since those times.

Despite the progress made in recent years, many questions concerning the tectonic evolution of the southwestern continental margin of Gondwana in Paleozoic times still remain. Some of these questions that could potentially be answered with further paleomagnetic investigations include: Which were the kinematics for the transfer of the Argentine Precordillera from Laurentia to Gondwana? Was the Puna Oriental–Famatina terrane a single (or multiple) volcanic arc? Were the Antofalla and Arequipa blocks separate terranes in the early Paleozoic? If so, what was the evolution of Arequipa? What was the paleogeographic evolution of Chilenia before its collision with Gondwana? Were some of these terranes related prior to their accretion to Gondwana? Was Patagonia a single tectonic block in Paleozoic times and what was its paleogeographic evolution before the Devonian?

Many more unanswered questions remain. It is hoped that further paleomagnetic studies in southern South America will yield some answers in the coming years.

ACKNOWLEDGMENTS

This work is the product of several years of paleomagnetic study. Collaboration and discussion with V. Ramos, J. Vilas, D. Tarling, S. Flint, P. Turner, B. Coira, M. Koukharsky, C. Vasquez, O. Bordonaro, and many colleagues at the Laboratorio Daniel Valencio in Buenos Aires are gratefully acknowledged. The final manuscript benefited from the critical reviews by Bob Butler and Rubén Somoza. The University of Buenos Aires, (particularly, UBACyT grants EX-135 and EX-002 supported the final stages of the study, Consejo Nacional de Investigaciones Científicas y Técnicas (Argentina), Fundación Antorchas, and the University of Plymouth (United Kingdom), contributed different kinds of support. This is a contribution to the International Geological Correlation Program Project 376 (Laurentia-Gondwana Connections before Pangea).

REFERENCES CITED

Aceñolaza, F. G., 1992, El sistema Ordovícico de Latinoamérica, *in* Gutierrez Marco, J. C., Saavedra, J., and Rabano. I., eds., El Paleozoico Inferior de Ibero-América: Public: Madrid, Universidad de Extremadura, Spain, p. 85–118.

Aitchenson, S. J., Harmon, R. S., Moorbath, S., Schneider, A., Soler, P., Soria-Escalante, E., Steele, G., Swainbank, I., and Wörner, G., 1995, Pb isotopes define basement domains of the Altiplano, central Andes: Geology, v. 23, p. 555–558.

Allmendinger, R. W., Ramos, V. A., Jordan, T. E., Palma, M., and Isacks, B. L., 1983, Paleogeography and Andean structural geometry, NW Argentina: Tectonics, v. 2, no. 1, p. 1–16.

Allmendinger, R. W., Figueroa, D., Snyder, D., Beer, J., Mpodozis, C. and Isacks, B. L., 1990, Foreland shortening and crustal balancing in the Andes at 30°S latitude: Tectonics, v. 9, no. 4, p. 789–809.

Astini, R. A., and Benedetto, J. L., 1996, Paleoenvironmental features and basin evolution of a complex volcanic-arc region in the pre-Andean western Gondwana: The Famatina belt: 3rd International Symposium on Andean Geodynamics, Extended Abstracts, p. 755–758.

Astini, R. A., and Vaccari, N. E., 1996, Sucesión evaporítica del Cámbrico inferior de la Precordillera: significado geológico: Revista de la Asociación Geológica Argentina, v. 51, p. 97–106.

Astini, R. A., Benedetto, J. L., and Vaccari, N. E., 1995, The early Paleozoic evolution of the Argentine Precordillera as a Laurentian rifted, drifted and collided terrane: a geodynamic model: Geological Society of America Bulletin, v. 107, p. 253–273.

Bahlburg, H., and Hervé, F., 1997, Geodynamic evolution and tectonostratigraphic terranes of northwestern Argentina and northern Chile: Geological Society of America Bulletin, v. 109, p. 869–884.

Baldis, B. A., and Bordonaro, O., 1981, Evolución de facies carbonáticas en la cuenca cámbrica de la Precordillera de San Juan: Actas 8º Congreso Geológico Argentino, San Luis, v. 1, p. 385–397.

Basei, M. A., Charrier, R., and Hervé, F., 1996, New ages (U-Pb, Rb-Sr, K-Ar) from supposed pre-Cambrian units in northern Chile: 3rd International Symposium on Andean Geodynamics, Extended Abstracts, p. 763–766.

Beck, M. E., Jr., 1989, Paleomagnetism of continental North America, implications for displacement of crustal blocks within the Western Cordillera, Baja California to British Columbia: Geological Society of America Bul-

Downloaded from specialpapers.gsapubs.org on June 2, 2012

letin, v. 101, p. 471–492.

Beck, M. E., García, A., Burmester, R. F., Munizaga, F., Hervé, F., and Drake, R., 1991, Paleomagnetism and geochronology of late Paleozoic granitic rocks from the Lake District of southern Chile: implications for accretionary tectonics: Geology, v. 19, p. 332–335.

Benedetto, J. L., and Sánchez, T. M., 1996, Paleobiogeography of brachiopod and molluscan faunas along the South American margin of Gondwana during the Ordovician, *in* Baldis, B. A., and Aceñolaza, F. G., eds., Early Paleozoic evolution in NW Gondwana: San Miguel de Tucumán, Argentina, Serie Correlación Geológica, v. 12, p. 23–38.

Benedetto, J. L., Vaccari, N. E., Carrera, M. G., and Sánchez, T. M., 1995, The evolution of faunal provincialism in the Argentine Precordillera during the Ordovician: new evidence and paleontological implications, *in* Cooper, J. D., Droser, M. L., and Finney, S. C., eds., Ordovician odyssey: SEPM Pacific Section, v. 77, p. 181–184.

Berquó, T. S., and Ernesto, M., 1997, Paleomagnetism of the early Paleozoic sediments from the Caacupé Group, eastern Paraguay: 8th Scientific Assembly, Uppsala, International Association of Geomagnetism and Aeronomy (IAGA 97), Abstracts, p. 59.

Blasco, G., Villar, L., and Zappettini, E. O., 1996, El complejo ofiolítico desmembrado de la Puna argentina. Provincias de Jujuy, Salta y Catamarca: Actas, 13º Congreso Geológico Argentino and 3º Congreso de Exploración de Hidrocarburos, Buenos Aires, v. 3, p. 653–667.

Bordonaro, O., Keller, M., and Lehnert, O., 1996, El Ordovícico de Ponón Trehué en la Provincia de Mendoza (Argentina): redefiniciones estratigráficas: Actas, 13º Congreso Geológico Argentino and 3º Congreso de Exploración de Hidrocarburos, Buenos Aires, v. 1, p. 541–550.

Borrello, A. V., 1965, Sobre la presencia del Cámbrico Inferior olenellidiano en la Sierra de Zonda, Precordillera de San Juan: Ameghiniana, v. 3, p. 313–318.

Buggisch, W., and Astini, R. A., 1993, The late Ordovician Ice age: new evidence from the Argentine Precordillera, *in* Findlay, R. H., Unrug, R., Banks, M. R., and Veevers, J. J., eds., Gondwana Eight; Assembly, evolution and dispersal: Rotterdam, Netherlands, Balkema, p. 439–447.

Cobbold, P. R., Gapais, D., and Rossello, E. A., 1991, Partitioning of transpressive motions within a sigmoidal foldbelt: the Variscan Sierras Australes, Argentina: Journal of Structural Geology, v. 13, no. 7, p. 743–758.

Coira, B., 1973, Resultados preliminares sobre la petrología del ciclo eruptivo concomitante con la sedimentación de la Formación Acoyte en la zona de Abra Pampa, prov. de Jujuy, Revista de la Asociación Geológica Argentina, v. 27, no. 1, p. 85–90.

Coira, B., and Koukharsky, M., 1991, Lavas en almohadilla, Ordovícico en el Cordón de Escaya, Puna Septentrional, Argentina: Actas, 6º Congreso Geológico Chileno, v. 1, p. 674.

Coira, B., Davidson, J., Mpodozis, C., and Ramos, V., 1982, Tectonic and magmatic evolution of the Andes of northern Argentina and Chile: Earth Science Reviews, v. 18, p. 89–95.

Conti, C. M., 1995, Paleomagnetismo del Paleozoico Inferior y Medio del Noroeste Argentino y sus implicancias en la evolución tectónica y geodinámica del márgen oeste del Gondwana [Ph.D. thesis]: Buenos Aires, Argentina, Universidad de Buenos Aires, 301 p.

Conti, C. M., Rapalini, A. E., and Vilas, J. F., 1995, Palaeomagnetism of the Silurian Lipeon Formation, NW Argentina, and the Gondwana apparent polar wander path: Geophysical Journal International, v. 121, p. 848–862.

Conti, C. M., Rapalini, A. E., Coira, B., and Koukharsky, M., 1996, Paleomagnetic evidence of an early Paleozoic rotated terrane in NW Argentina. A clue for Gondwana-Laurentia interaction?: Geology, v. 24, p. 953–956

Dalla Salda, L., Cingolani, C., and Varela, R., 1992, Early Paleozoic orogenic belt of the Andes in southwestern South America: result of Laurentia-Gondwana collision?: Geology, v. 20, p. 617–620.

Dalmayrac, B., Lancelot, J. R., and Leyreloup, A., 1977, Two billion year granulites in the late Precambrian metamorphic basement along the southern Peruvian coast: Science, v. 198, p. 49–51.

Dalmayrac, B., Laubacher, G., Marocco, R., Martinez C., and Tomasi, P., 1980, La Chaine Hercynienne d'Amerique du Sud. Structure et evolution d'une orogene intracratonique: Geologische Rundschau, v. 69, no. 1, p. 1–21.

Dalziel, I. W. D., 1997, Neoproterozoic-Paleozoic geography and tectonics: review, hypothesis, environmental speculation: Geological Society of America Bulletin, v. 109, p. 16–42.

Dalziel, I. W. D., Dalla Salda, L. H., Cingolani C., and Palmer, A. R., 1996, The Argentine Precordillera, a Laurentian terrane; Penrose Conference Report: GSA Today, February, p. 16–18.

Damm, K. W., Pichowiak, S., Harmon, R. S., Todt, W., Keley, S., Omarini, R., and Niemeyer, H., 1990, Pre-Mesozoic evolution of the central Andes: the basement revisited, *in* Mahlburgh Kay, S., and Rapela, C. W., eds., Plutonism from Antarctica to Alaska: Geological Society of America Special Paper, v. 241, p. 101–126.

de Wit, M., Jeffery, M., Bergh, H., and Nicolaysen, L., 1988, Geological map of Gondwana, with explanations and references: Tulsa, Oklahoma, American Association of Petroleum Geologists, scale 1:10,000,000.

Forsythe, R. D., and Mpodozis, C., 1979, El archipiélago Madre de Dios, Patagonia Occidental, Magallanes: rasgos generales de la estratigrafía y estructura del "basamento" pre-Jurásico Superior: Revista Geológica de Chile, v. 7, p. 13–29.

Forsythe, R. D., Kent, D. V., Mpodozis, C., and Davidson, J., 1987, Palaeomagnetism of Permian and Triassic rocks, central Chilean Andes, *in* Elliot, D. H., Collinson, J. W., and McKenzie, G. D., eds., Gondwana Six: American Geophysical Union Geophysical Monograph Series, p. 241–252.

Forsythe, R. D., Davidson, J., Mpodozis, C., and Jesinkey, C., 1993, Lower Paleozoic relative motion of the Arequipa block and Gondwana; paleomagnetic evidence from Sierra de Almeida of northern Chile: Tectonics, v. 12, p. 219–236.

Götze, H. J., Lahmeyer, B., Schmidt, S., and Strunk, S., 1994, The lithospheric structure of the central Andes (20–26°S) as inferred from interpretation of regional gravity, *in* Reutter, K. J., Scheuber, E., and Wigger, P. J., eds., Tectonics of the southern central Andes: Berlin, Springer-Verlag, p. 7–21.

Grunow, A. M., 1995, Implications for Gondwana of new Ordovician paleomagnetic data from igneous rocks in southern Victoria Land, east Antarctica: Journal of Geophysical Research, v. 100, p. 12589–12603.

Japas, M. S., 1987, Análisis cuantitativo de la deformación en el Sector Oriental de las Sierras Australes de Buenos Aires y su implicancia geodinámica [Ph.D. thesis]: Buenos Aires, Argentina, Universidad de Buenos Aires, 359 p.

Jordan, T. E., Isacks, B. L., Ramos V. A., and Allmendinger, R. W., 1983, Mountain building in the central Andes: Episodes, v. 3, p. 20–26.

Keidel, J., 1925, Sobre el desarrollo paleogeográfico de las grandes unidades geológicas de la Argentina: Sociedad Argentina de Estratigrafía y Geografía, GAEA, Anales, v. 4, p. 251–312.

Koukharsky, M., and Mirré, J. C., 1974, Nuevas evidencias del vulcanismo ordovícico en la Puna: Revista de la Asociación Geológica Argentina, v. 29, no. 1, p. 128–134.

Lopez, J., and Toselli, A., 1993, La faja milonítica de TIPA: faldeo oriental del Sistema de Famatina, Argentina, Actas, 12º Congreso Geológico Argentino and 2º Congreso de Exploración de Hidrocarburos, Mendoza, v. 3, p. 39–42.

Lork, A., and Bahlburgh, H., 1993, Precise U-Pb ages of monazites from the Faja Eruptiva de la Puna Oriental and the Cordillera Oriental, NW Argentina: Actas, 12º Congreso Geológico Argentino and 2º Congreso de Exploración de Hidrocarburos, Mendoza, v. 4, p. 1–6.

Mac Niocaill, C., van der Pluijm, B. A., and Van der Voo, R., 1997, Ordovician paleogeography and the evolution of the Iapetus ocean: Geology, v. 25, p. 159–162.

Mahlburg Kay, S., Ramos, V. A., Mpodozis, C., and Sruoga, P., 1989, Late Paleozoic to Jurassic silicic magmatism at the Gondwana margin: analogy to the Middle Proterozoic in North America?: Geology, v. 17, p. 324–328.

Mahlburg Kay, S., Orrell, S. and Abbruzzi, J. M., 1996, Zircon and whole rock Nd-Pb evidence for a Grenville age and a Laurentian origin for the basement of the Precordillera in Argentina: Journal of Geology, v. 104, p. 637–648.

Mannheim, R., 1993, Génesis de las volcanitas eopaleozoicas del Sistema de Famatina, Actas, 12º Congreso Geológico Argentino and 2º Congreso de Exploración de Hidrocarburos, Mendoza, v. 4, p. 147–155.

Downloaded from specialpapers.gsapubs.org on June 2, 2012

Martinez, C., 1980, Structure et evolution de la Chaine Hercynienne et de la Chaine Andine dans le nord de la Cordillere des Andes de Bolivie: Travaux et Documents de L'Ostrom, no. 119, 352 p.

Meert, J. G., and Van der Voo, R., 1996, Paleomagnetic and $^{40}AR/^{39}Ar$ study of the Sinyai Dolerite, Kenya: implications for Gondwana Assembly: Journal of Geology, v. 104, p. 131–142.

Mon, R., and Hongn, F., 1987, Estructura del Ordovícico de la Puna: Revista de la Asociación Geológica Argentina, v. 31, p. 65–72.

Moya, M., Malanca, S., Hongn, F. and Bahlburgh, H., 1993, El Tremadoc temprano en la Puna Occidental Argentina: Actas, 12º Congreso Geológico Argentino and 2º Congreso de Exploración de Hidrocarburos; Mendoza, v. 2, p. 20–30.

Mpodozis, C., and Ramos, V. A., 1989, The Andes of Chile and Argentina, *in* Ericksen, G. E., Cañas Pinochet, M. T., and Reinemud, J. A., eds., Geology of the Andes and its relation to hydrocarbon and mineral resources: Circumpacific Council for Energy and Mineral Resources Earth Sciences Series, v. 11, p. 59–90.

Omarini, R., Viramonte, J., Cordani, U., Salfity, J., and Kawashita, K., 1984, Estudio geocronológico Rb-Sr de la Faja Eruptiva de la Puna en el sector de San Antonio de los Cobres, Provincia de Salta: Actas, 9º Congreso Geológico Argentino, S. C. Bariloche, v. 3, p. 146–158.

Pacci, D., Munizaga, F., Hervé, F., Kawashita, K., and Cordani, U., 1980, Acerca de la edad Rb-Sr precámbrica de rocas de la Formación Esquistos de Belén, Departamento de Parinacota, Chile: Revista Geológica de Chile, v. 11, p. 43–58.

Palma, M. A., Parica, P. D., and Ramos, V. A., 1986, El granito de Archibarca: su edad y significado tectónico, provincia de Catamarca: Revista de la Asociación Geológica Argentina, v. 41, p. 414–419.

Poulsen, V., 1958, Contribution to the Middle Cambrian paleontology and stratigraphy of Argentina: Copenhague Matematisc fysiske Meddelelser, Der kongelige Danske Videnskabernes Selsbak, v. 31, p. 1–22.

Ramos, V. A., 1984, Patagonia: un continente paleozoico a la deriva? Actas, 9º Congreso Geológico Argentino, S. C. Bariloche, v. 2, p. 311–325.

Ramos, V. A., 1986, El diastrofismo Oclóyico: un ejemplo de tectónica de colisión durante el Eopaleozoico en el Noroeste Argentino: Revista del Instituto Geológico Minero, Jujuy, v. 6, p. 13–28.

Ramos, V. A., 1988, Late Proterozoic–early Paleozoic of South America—a collisional history: Episodes, v. 11, no. 3, p. 168–174.

Ramos, V. A., Jordan, T. E., Allmendinger, R. W., Mpodozis, C., Kay, S. M., Cortés J. M., and Palma, M., 1986, Paleozoic terranes of the central Argentine—Chilean Andes: Tectonics, v. 5, no. 6, p. 855–880.

Rapalini, A. E., 1998, Syntectonic magnetization of the mid-Paleozoic Sierra Grande Formation. Further constraints for the tectonic evolution of Patagonia: Journal of the Geological Society of London, v. 155, p. 105–114.

Rapalini, A. E., and Astini, R. A., 1998, Paleomagnetic confirmation of the Laurentian origin of the Argentine Precordillera: Earth and Planetary Science Letters, v. 155, p. 1–14.

Rapalini, A. E., and Tarling, D. H., 1993, Multiple magnetizations in the Cambro-Ordovician carbonate platform of the Argentine Precordillera and their tectonic implications: Tectonophysics, v. 227, p. 49–62.

Rapalini, A. E., and Vilas, J.F., 1991, Preliminary paleomagnetic data from the Sierra Grande Formation: tectonic consequences of the first Middle Paleozoic paleopoles from Patagonia: Journal of South American Earth Sciences, v. 4, no. 1–2, p. 25–41.

Rapalini, A. E., Vilas, J. F., Bobbio, M. L., and Valencio, D. A., 1989, Geodynamic interpretations from paleomagnetic data of late Paleozoic rocks in the southern Andes, *in* Hillhouse, J. W., ed., Deep structure and past kinematics of accreted terranes: American Geophysical Union Geophysical Monograph Series, v. 5, no. 50, p. 41–57.

Rapalini, A. E., Tarling, D. H., Turner, P., Flint, S., and Vilas, J. F., 1994, Palaeomagnetism of the Carboniferous Tepuel Group, central Patagonia, Argentina: Tectonics, v. 13, no. 5, p. 1277–1294.

Rapela, C. W., Coira, B., Toselli, A., and Saavedra, J., 1992, El magmatismo del Paleozoico Inferior en el sudoeste de Gondwana, *in* Gutierrez Marco, J. C., Saavedra, J., and Rabano. I., eds., El Paleozoico Inferior de Ibero-América: Madrid, Publicacion Universidad de Extremadura, Spain, p. 21–68.

Rogers, J. J. W., Unrug, R., and Sultan, M., 1994, Tectonic assembly of Gondwana: Journal of Geodynamics, v. 19, p. 1–34.

Sellés Martinez, J., 1988, The relationship between Patagonian and South American blocks: frontal collision or transpression?: 7th Gondwana Symposium, Sao Paulo, Brazil, Abstracts, p. 139.

Sempere, T., 1995, Phanerozoic evolution of Bolivia and adjacent regions, *in* Tankard, A. J., Suárez Soruco, R., and Welsink, H. J., eds., Petroleum basins of South America: American Association of Petroleum Geologists Memoir, v. 62, p. 207–230.

Shackleton, R. M., Ries, A. C., Coward, M. P., and Cobbold, P. R., 1979, Structure, metamorphism and geochronology of the Arequipa Massif of coastal Perú: Geological Society of London Memoir, v. 12, p. 1–21.

Thomas, W., and Astini, R. A., 1996, The Argentine Precordillera: a traveler from the Ouachita embayment of North American Laurentia: Science, v. 273, p. 752–757.

Tomezzoli, R. N., and Vilas, J .F., 1996, Paleomagnetismo del Grupo Pillahiuncó en Sierra de la Ventana (Estancias Las Julianas y San Carlos): Actas, 13º Congreso Geológico Argentino and 3º Congreso de Exploración de Hidrocarburos, Buenos Aires, v. 2, p. 481–488.

Torsvik, T., Smethurst, M., Meert, J., Van der Voo, R., McKerrow, S., Brasier, M., Sturt, B., and Walderhaug, H., 1996, Continental break-up and collision in the Neoproterozoic and Paleozoic—a tale of Baltica and Laurentia: Earth Science Review, v. 40, p. 229–258.

Tosdal, R. M., Munizaga, F., Williams, W. C., and Bettencourt, J. S., 1994, Middle Proterozoic crystalline basement in the central Andes, western Bolivia and northern Chile: a U-Pb and Pb isotopic perspective: Actas, 7º Congreso Geológico Chileno, v. 2, p. 1464–1467.

Toselli, A. J., Durand, F. R., Rossi de Toselli, J. N., and Saavedra, J., 1996, Esquema de evolución geotectónica y magmática eopaleozoica del Sistema de Famatina y sectores de Sierras Pampeanas: Actas, 13º Congreso Geológico Argentino and 3º Congreso de Exploración de Hidrocarburos, Buenos Aires, v. 5, p. 443–462.

Truco, S., and Rapalini, A. E., 1996, New evidence of a widespread Permian remagnetizing event in the central Andean zone of Argentina: 3rd International Symposium on Andean Geodynamics, St. Malo, France, Extended Abstracts, p. 799–802.

Vaccari, N. E., 1990, Primer hallazgo de trilobites del Cámbrico inferior en la Provincia de la Rioja (Precordillera Septentrional): Revista de la Asociación Geológica Argentina, v. 43, p. 558–561.

Vaccari, N. E., Benedetto, J. L., Waisfeld, B. G., and Sánchez, T., 1992, La fauna de *Neseuretus* en la Formación Suri (oeste de Argentina): edad y relaciones paleobiogeográficas: Revista Española de Paleontología, v. 8, no. 2, p. 185–190.

Valencio, D. A., Vilas, J. F., and Mendia, J. E., 1980, Palaeomagnetism and K-Ar ages of Lower Ordovician and Upper Silurian–Lower Devonian rocks from north-west Argentina: Geophysical Journal of the Royal Astronomical Society, v. 62, p. 27–39.

Van der Voo, R., 1993, Paleomagnetism of the Atlantic, Tethys and Iapetus Oceans: Cambridge, United Kingdom, Cambridge University Press, 411 p.

Van der Voo, R, 1994, True polar wander during the middle Paleozoic?: Earth and Planetary Science Letters, v. 122, p. 239–243.

Varela R., Dalla Salda, L., Cingolani, C., and Gomez, V., 1991, Estructura, petrología y geocronología del basamento de la región del Limay, provincias de Río Negro y Neuquén, Argentina: Revista Geológica de Chile, v. 18, p. 147–163.

von Gosen, W., Buggisch, W., and Dimieri, L. V., 1990, Structural and metamorphic evolution of the Sierras Australes (Buenos Aires Province/Argentina): Geologische Rundschau, v. 79, no. 3, p. 797–821.

Windhausen, A., 1931, Geología Argentina. II parte. Geología Histórica y Regional del territorio argentino: Buenos Aires, Argentina. Editorial Peusser, 645 p.

Wörner, G., Moorbath, S., and Harmon, R. S., 1992, Andean Cenozoic volcanic centers reflect basement isotopic domains: Geology, v. 20, p. 1103–1106.

MANUSCRIPT ACCEPTED BY THE SOCIETY SEPTEMBER 4, 1998